

Allen Tannenbaum

(January 25, 1953 - December 28, 2023)

celebrating the life and legacy,
of our dearest friend and colleague, my brother!





Gainesville 1983



Infectious smile, exuberance, optimism
loyal, dedicated, caring for everyone
outpouring creativity and humor



Athens 2015



Miami 2018



Minneapolis 1993



Cairnes, Australia 2001



New York 2023



Minneapolis 1992



Brown 2023

1970's - early 1980's



PROCEEDINGS OF THE
AMERICAN MATHEMATICAL SOCIETY
Volume 68, Number 1, January 1978

DEGENERATIONS OF CURVES IN $\mathbb{P}^3$

ALLEN TANNENBAUM

ABSTRACT. In this paper we prove every connected, reduced curve in $\mathbb{P}^3$ of arithmetic genus 0, may be flatly smoothed. Moreover, we give a new example of a reduced singular curve in $\mathbb{P}^3$ which cannot be flatly smoothed.

Columbia University, B.A. in Mathematics, June 1973.

Harvard University, Ph.D. in Mathematics, June 1976.

Ph.D. advisor: Professor Heisuke Hironaka, Harvard University.

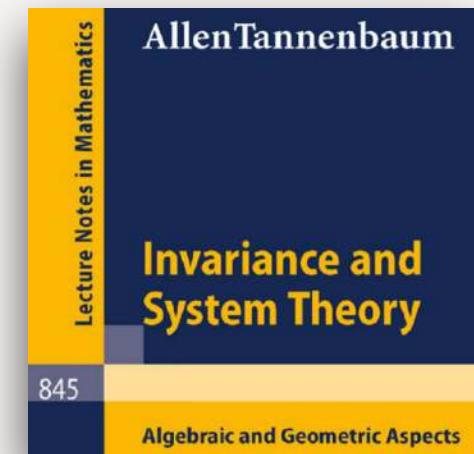
Thesis title: *Deformations of 1-Cycles and the Chow Scheme.*



Allen Tannenbaum was born in New York City in 1953. He received the Ph.D. degree in mathematics from Harvard University, Cambridge, MA, in 1976.

At present he is an Associate Professor of Mathematics at Ben-Gurion University of Neger, Beer Sheva, Israel. His research interests are in the robust control of linear systems, algebraic system theory, and algebraic geometry.

1970's - early 1980's



1980 - the beginnings of Robust Control

Feedback stabilization of linear dynamical plants with uncertainty in the gain factor

ALLEN TANNENBAUM

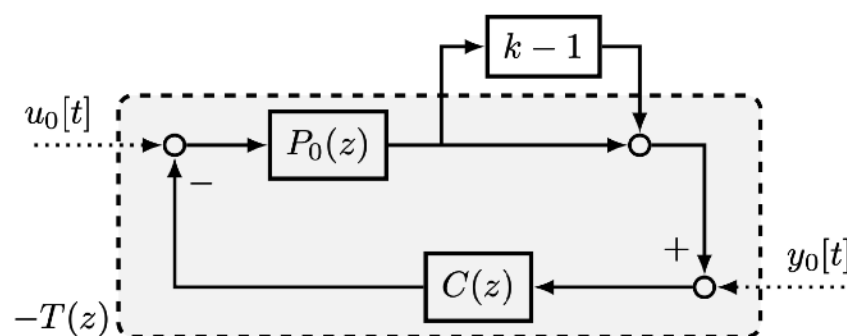


Fig. 2. Feedback control system under gain uncertainty.

$$T(z) := \frac{P_0(z)C(z)}{1 + P_0(z)C(z)}.$$

must exclude values in

$$\mathcal{S}_{k_1, k_2} := \left\{ \zeta \in \mathbb{R} : \zeta \leq \frac{-1}{k_2 - 1} \right\} \cup \left\{ \zeta \in \mathbb{R} : \zeta \geq \frac{-1}{k_1 - 1} \right\}.$$

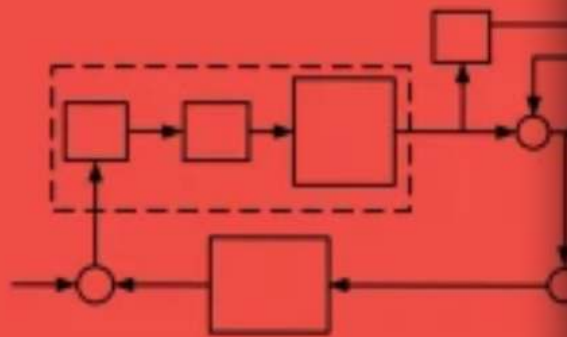
- A. Tannenbaum, "Feedback stabilization of plants with uncertainty in the gain factor," *Int. J. Contr.*, vol. 32, pp. 1-16, 1980.
- A. Tannenbaum, *Invariance and System Theory: Algebraic and Geometric Aspects*. New York: Springer-Verlag, 1981.
- , "Modified Nevanlinna-Pick interpolation and feedback in the gain factor," *Int. J. Contr.*, vol. 36, pp. 331-336, 1982.

First use of Nevanlinna-Pick theory for feedback design
First solution of an optimal robust control problem 1980

C. Foias, H. Özbay and
A. Tannenbaum

Robust Control of Infinite Dimensional Systems

Frequency Domain Methods



tes in Mathematics

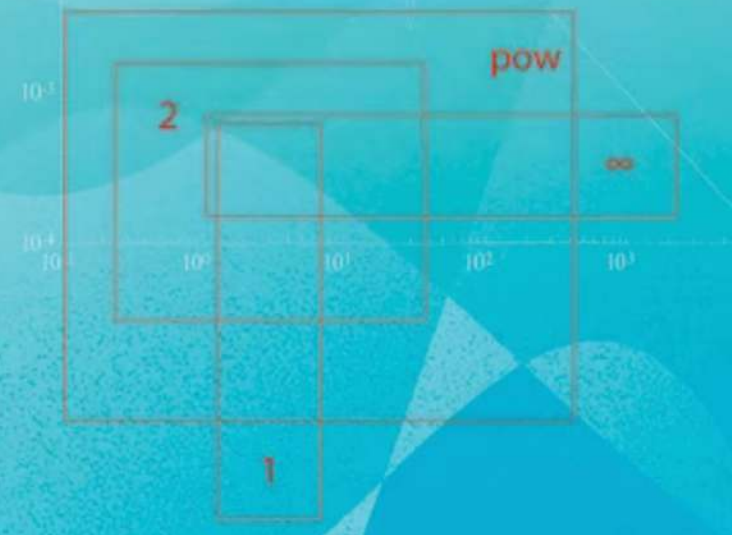
Allen Tannenbaum

Invariance and System Theory

Algebraic and Geometric Aspects

FEEDBACK CONTROL THEORY

John C. Doyle
Bruce A. Francis
Allen R. Tannenbaum



1990's -

Main Fields of Interest

Image processing, computer vision, mathematical systems theory, control, medical imaging, computer graphics, control of semiconductor fabrication processes, robotics, bioinformatics, operator theory, functional analysis, algebraic geometry, differential geometry, invariant theory, and partial differential equations.

multi-scale resolution

denoising

active contours

image skeletonization

invariant heat equation

group invariances in PDE's

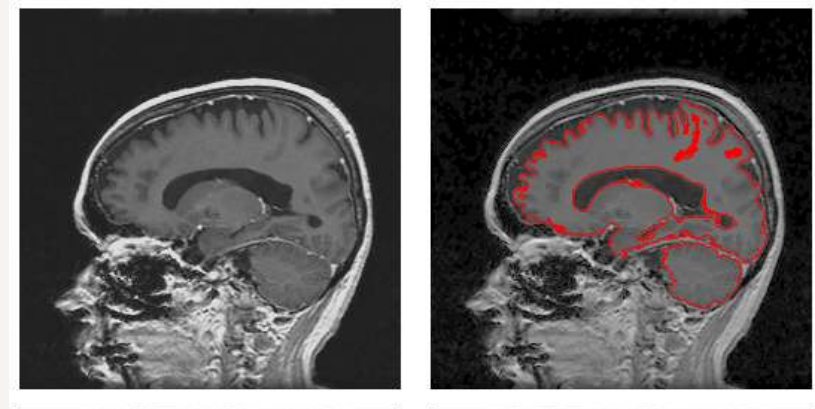
surface flattening

object recognition

control with visual feedback

...

Biomedical imaging



The fundamental equation of image processing

Equi-affine Curve Shortening

שליש kappa

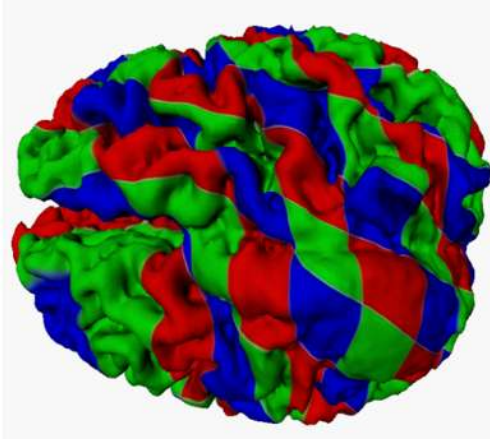
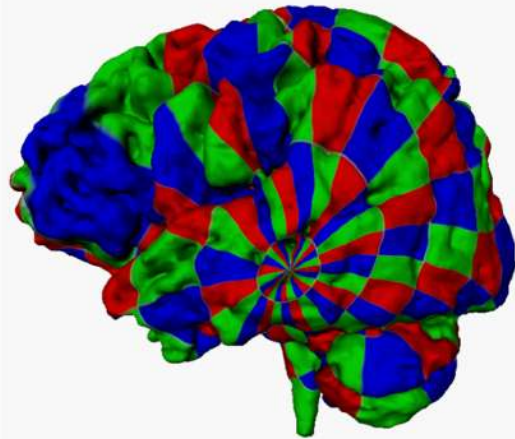
$$C_t = -\sqrt[3]{\kappa} \mathbf{N}$$

$$u_t = \sqrt[3]{u_{xx}}$$

$$\Phi_t = (\Phi_y^2 \Phi_{xx} - 2\Phi_x \Phi_y \Phi_{xy} + \Phi_x^2 \Phi_{yy})^{1/3}$$

with Guillermo Sapiro

Coordinate System on Cortical Surface



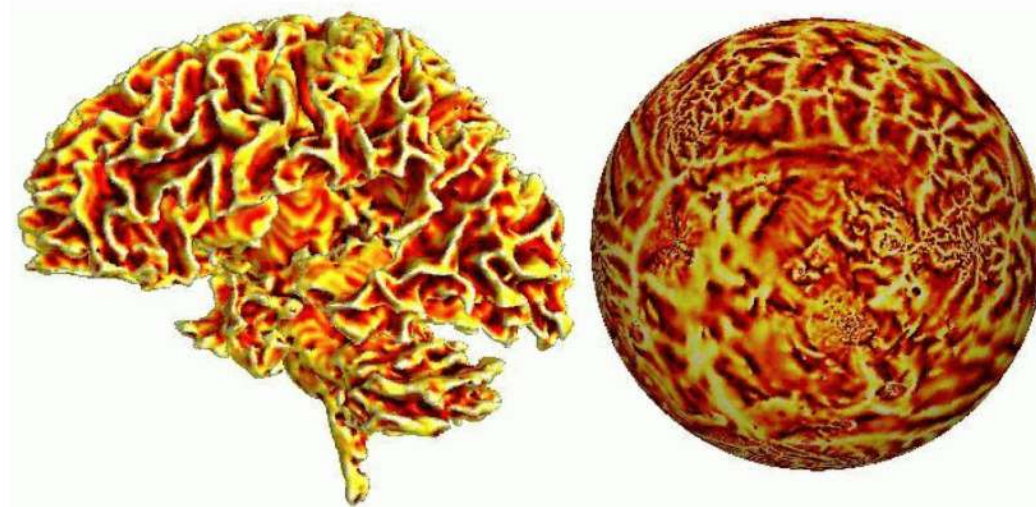
IEEE TRANSACTIONS ON MEDICAL IMAGING, VOL. 18, NO. 8, AUGUST 1999

On the Laplace–Beltrami Operator and Brain Surface Flattening

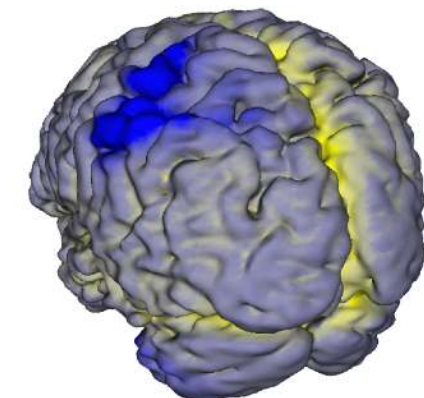
Sigurd Angenent, Steven Haker, Allen Tannenbaum,* and Ron Kikinis

$$A(t) := \int_0^1 \int_0^1 (\|S_u\|^2 \|S_v\|^2 - \langle S_u, S_v \rangle^2)^{1/2} du dv.$$

White Matter Segmentation and Flattening



A huge
body of work

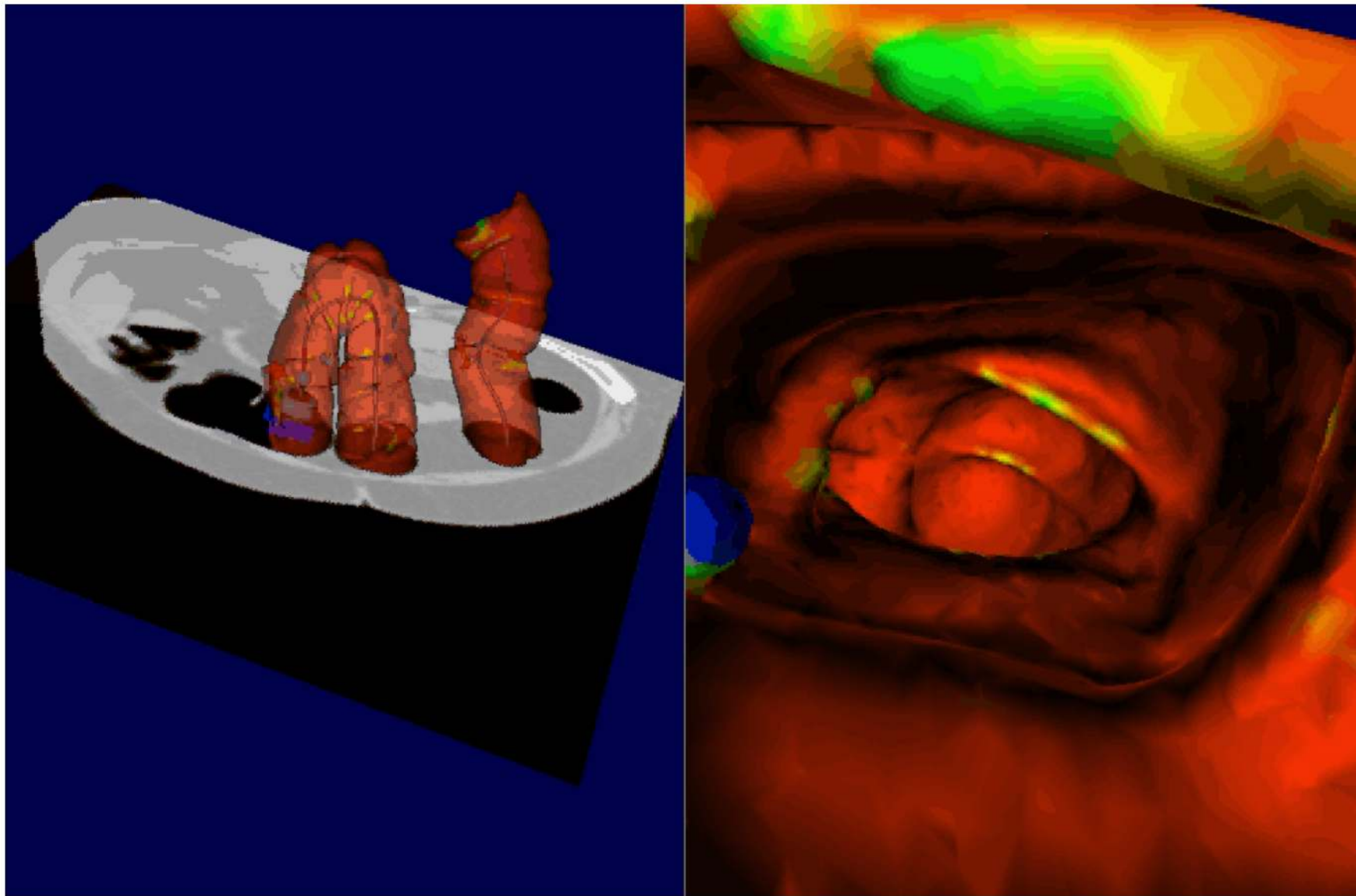


non-invasive methods

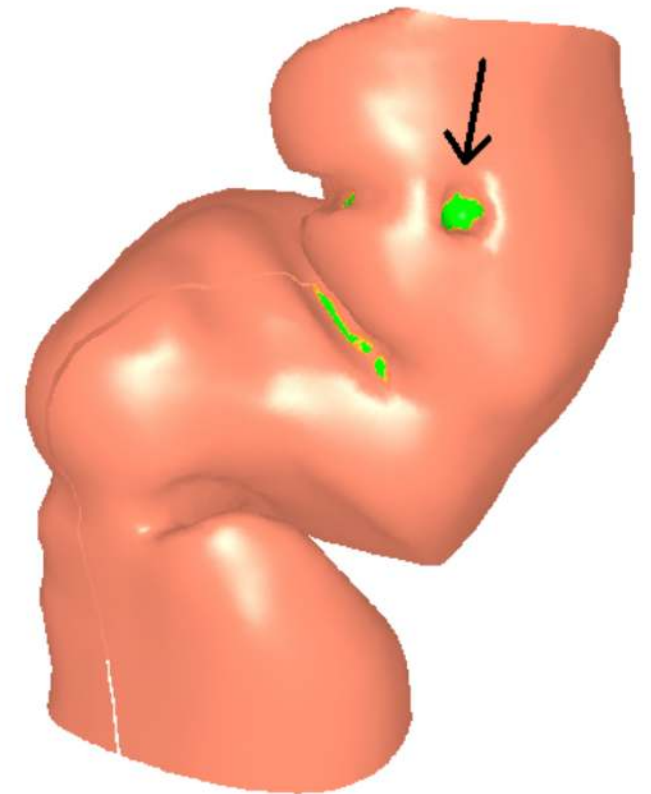
colon fly-through!

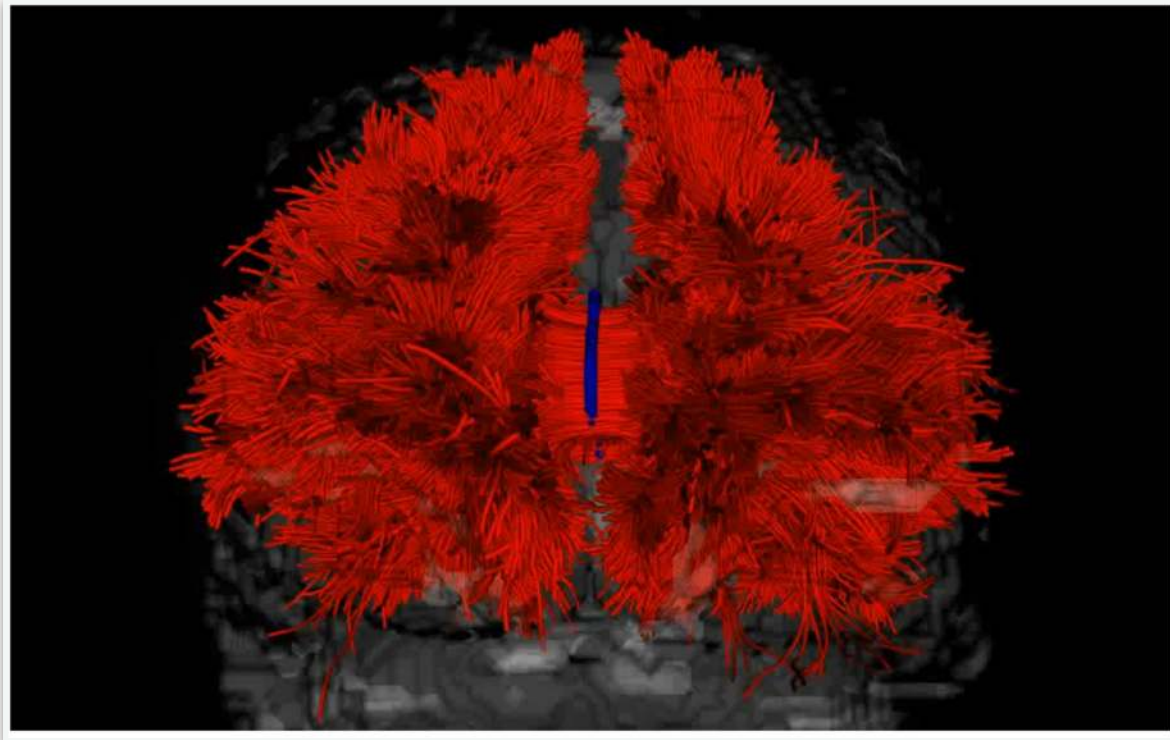
virtual colonoscopy

Path-Planning Deluxe



Polyp Highlighted



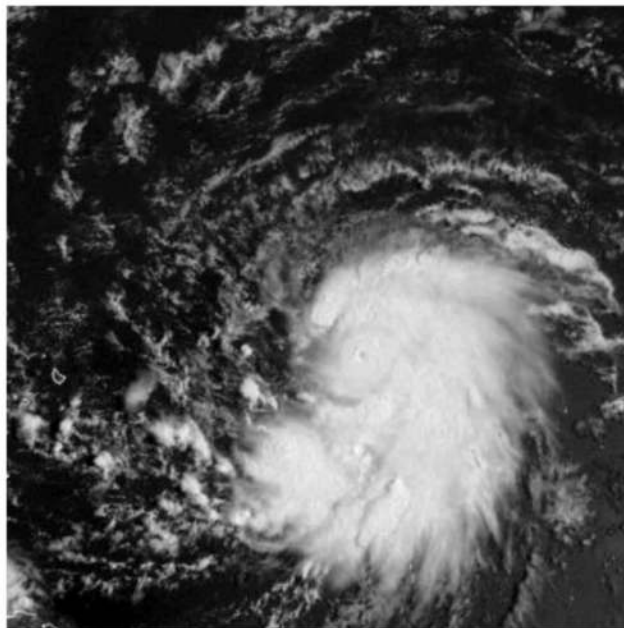


Optimal Mass Transport

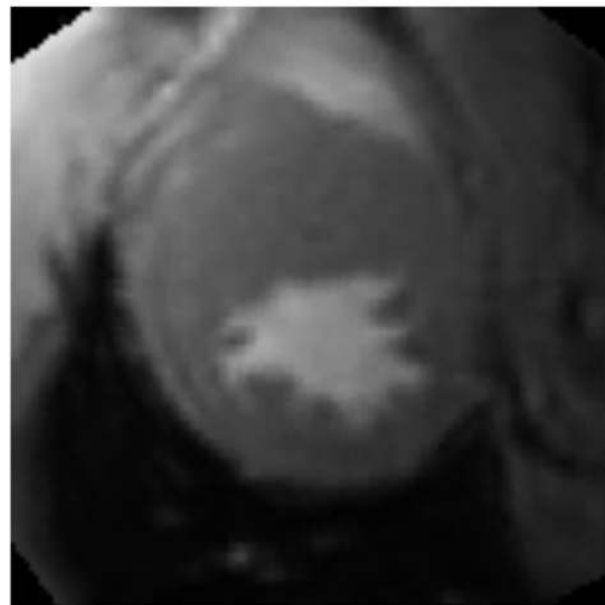
First gradient flow algorithm
Angenent-Haker-Tannenbaum 2003

Flame Motion

Hurricane Dean



Beating Heart

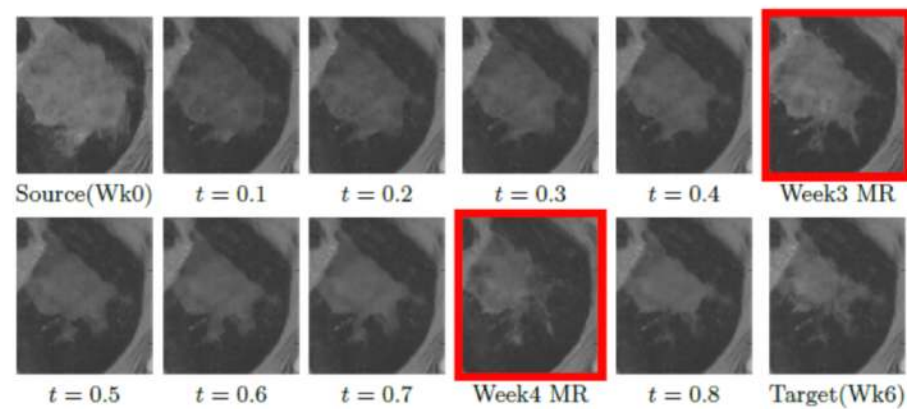


Optimal Mass Transport

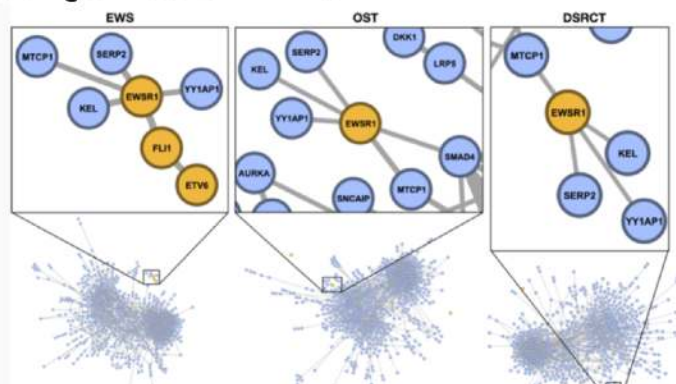
biomedical applications

Memorial Sloan Kettering Cancer Center

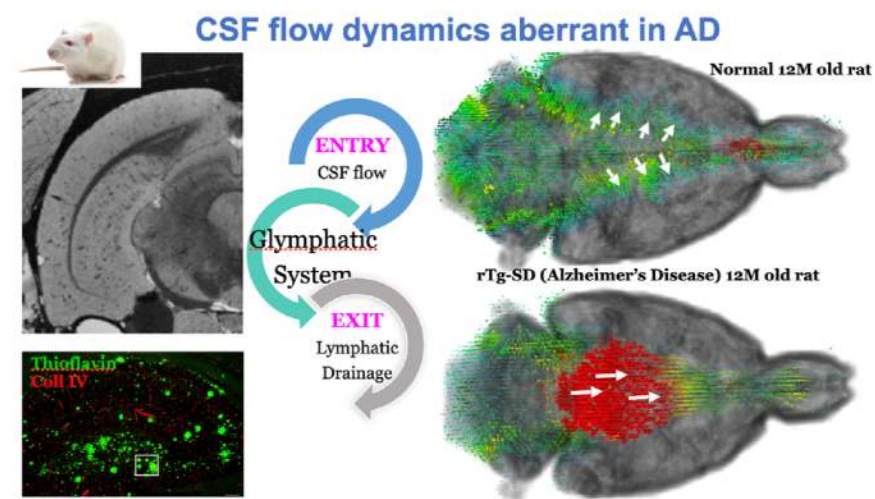
Data augmentation:
alternative to GAN (tumor)



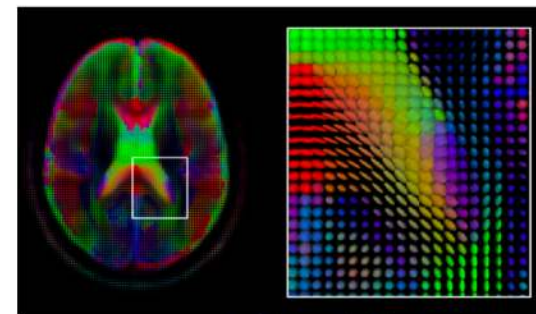
OMT & robustness: gene networks
pathways, hubs, drug response in
conjunction with ML



OMT flows: glymphatic system
neurodegenerative diseases



Matrix OMT:
DTI, Quantum information



...and many more



Allen Tannenbaum

Distinguished Professor, Computer Science/Applied Mathematics, [Stony Brook University](#)

Verified email at stonybrook.edu - [Homepage](#)

[Medical Imaging](#) [Systems and Control](#) [Computer Vision](#) [Applied Mathematics](#)
[Image Processing](#)

Harvard University 1973-1976

Weizmann Institute of Science 1976-1978, 1980-1983

Institute des Haute Études Scientifiques 1978

ETH Zürich 1978-1980

University of Florida 1982-1984

Ben-Gurion University 1984-1986

McGill University 1985-1986

University of Minnesota 1986-2002

Technion 1989-1992, 2005-2010

Georgia Tech 1999-2011

University of Alabama 2012-2013

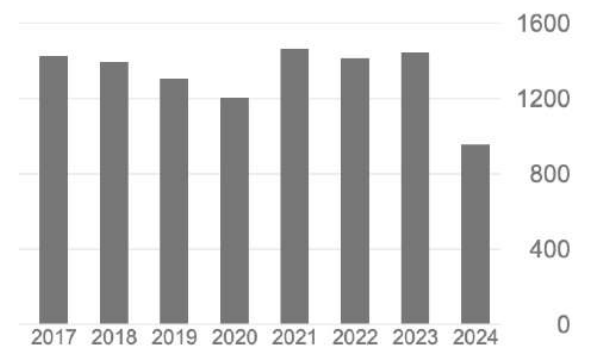
Stony Brook University 2013-2023

Memorial Sloan Kettering Cancer Center 2015-2023

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Lasting legacy:

algebraic geometry

mathematical systems theory

robust control theory

image processing

computer vision

gradient flow methods

optimal mass transport

and many more



To Allen, in loving memory